Fluid Power Graphic Symbols Reference

Standard: ANSI Y32.10 / ISO 1219
Course: Fluid Power Systems (MEC 431) — BUC
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1. General Symbol Rules

> **Purpose:** Symbols represent *function, flow paths, and actuation method* — NOT the internal mechanical design or physical dimensions of the component.

1.1 Lines and Conductors

Line Type	Appearance	Function
Working / Main Line	Solid continuous line	Carries pressurized fluid (supply and return)
Pilot Line	Long dashes (- - - -)	Control signal line (low pressure)
Drain / Exhaust Line	Short dashes or dots (· · · · ·)	Leakage or exhaust return to tank
Enclosure Outline	Dash-dot (- · - · -)	Boundary of a composite/manifold assembly
Flexible Line	Curved arc	Flexible hose connection
Lines Crossing	Lines cross without a dot	No connection (just crossing)
Lines Joining	Lines meet with a dot	Connected junction
Line with Fixed Restriction	X on the line	Fixed orifice / restriction

1.2 Arrows and Indicators

- **45° Arrow through a symbol:** Indicates **variable / adjustable** component.
- **Arrow parallel to valve square:** Indicates **pressure-compensated** component.

- **Filled triangle (▶):** Hydraulic flow (oil).
- **Open triangle (▷):** Pneumatic flow (air) or exhaust to atmosphere.

2. Fluid Conditioning & Storage

Component	Symbol Description	Function
Reservoir (Vented)	Open-top rectangular basin	Stores fluid, allows air breathing
Reservoir (Pressurized)	Closed-top rectangular basin	Stores fluid under pressure
Accumulator (Spring)	Oval capsule divided by spring symbol	Stores energy using spring compression
Accumulator (Gas-Charged)	Oval capsule divided by curved line	Stores energy using compressed gas (N_2)
Accumulator (Weighted)	Oval capsule with weight on top	Stores energy using gravitational weight
Filter / Strainer	Diamond with dashed vertical line inside	Removes solid contaminants from fluid
Cooler (Heat Exchanger)	Diamond with outward-pointing triangles	Removes heat from fluid
Heater	Diamond with inward-pointing triangles	Adds heat to fluid
FRL Unit (Pneumatic)	Filter + Regulator + Lubricator combined	Conditions compressed air before use

3. Actuators (Linear & Rotary)

3.1 Hydraulic Cylinders

Type	Description	Key Feature
Single-Acting (Spring Return)	One port, internal spring for retraction	Force in one direction only
Single-Acting (Gravity Return)	One port, external load retracts piston	No spring needed (weight returns piston)
Double-Acting (Single Rod)	Two ports, rod extends from one side	Force in both directions, unequal areas

Double-Acting (Double Rod)	Two ports, rods extend from both sides	Equal areas → equal force and speed both ways
Cushioned Cylinder	Adjustable orifice symbol at stroke end	Decelerates piston near end of stroke
Pressure Intensifier	Two connected pistons of different areas	$P_2 = P_1 \times \frac{A_1}{A_2}$ — multiplies pressure

3.2 Pumps and Motors

Component	Symbol	Description
Fixed-Displacement Hydraulic Pump	Circle + filled triangle pointing **outward**	Delivers constant flow per revolution
Variable-Displacement Hydraulic Pump	Same + 45° arrow through it	Flow output can be adjusted
Bi-Directional Pump	Circle + two filled triangles (opposite)	Can pump in either direction
Pressure-Compensated Variable Pump	Variable pump + parallel arrow	Auto-adjusts displacement based on pressure
Fixed-Displacement Hydraulic Motor	Circle + filled triangle pointing **inward**	Converts hydraulic energy to rotation
Variable-Displacement Motor	Same + 45° arrow through it	Speed/torque ratio can be adjusted
Pneumatic Compressor	Circle + open triangle pointing outward	Compresses air
Pneumatic Motor	Circle + open triangle pointing inward	Converts pneumatic energy to rotation

4. Actuating Mechanisms (Controls)

Type	Symbol Description	Application
Push Button (Manual)	Small circle on actuator side	Operator presses to shift valve
Lever (Manual)	Angled line from actuator side	Operator moves lever to shift valve
Foot Pedal	Angled line with pedal shape	Hands-free valve actuation
Roller (Mechanical)	Circle at end of line	Activated by cam or machine part in both directions
Idle Return Roller	Roller with one-way return bracket	Activated in one direction only, free return
Solenoid (Electrical)	Rectangular block on actuator side	Electrically shifted valve (most common in automation)
Pilot Pressure (Hydraulic/Pneumatic)	Dashed line to actuator side	Valve shifted by fluid pressure signal
Spring Return	Zigzag line on return side	Returns valve to default position when signal removed
Detent	Small notch symbol on actuator	Mechanically holds valve in last shifted position

5. Directional Control Valves (DCVs)

5.1 Naming Convention

Format: Number of Ports / Number of Positions

Examples:

- **2/2:** 2 ports, 2 positions (on/off valve)
- **3/2:** 3 ports, 2 positions (common in pneumatics)
- **4/2:** 4 ports, 2 positions (cylinder extend/retract, no neutral)

- **4/3:** 4 ports, 3 positions (cylinder extend/neutral/retract)

Port Labels: P = Pressure (supply), T = Tank (return), A & B = Work ports (to actuator)

5.2 Center Conditions for 4/3 Valves

> **Important:** The center condition determines system behavior when the valve is in neutral. This is a critical design choice.

Center Type	P	T	A	B	Effect
Closed Center	Blocked	Blocked	Blocked	Blocked	All ports blocked. Pump deadheads → needs relief valve. Load is locked.
Tandem Center	→ T	Open	Blocked	Blocked	Pump unloads to tank (low energy). Load is locked.
Open Center	→ T	Open	→ T	→ T	All ports connected to tank. Pump unloads. Load can drift (float).
Float Center	Blocked	Open	→ T	→ T	Pump blocked. A and B connected to tank. Load can float freely.

5.3 Other Valve Types

Valve	Normal State	Function
Check Valve (Non-Return)	Normally Closed	Allows flow in one direction only; blocks reverse flow
Pilot-Operated Check Valve	Normally Closed	Same as check valve but can be opened by external pilot signal
Shuttle Valve	—	Selects the higher pressure from two sources (OR logic)

6. Pressure & Flow Control Valves

6.1 Pressure Control Valves

Valve	Normal State	Senses Pressure At	Function
Relief Valve	Normally Closed (NC)	Inlet (upstream)	Limits maximum system pressure by diverting excess flow to tank. Acts as a safety device.
Pressure Reducing Valve	Normally Open (NO)	Outlet (downstream)	Maintains a constant, lower pressure in a secondary branch regardless of main system pressure.
Sequence Valve	Normally Closed (NC)	Inlet (upstream)	Opens to allow flow to a second actuator only after the first actuator has reached a set pressure.
Unloading Valve	Normally Closed (NC)	External pilot	Diverts pump flow to tank at low pressure when signaled by an external pilot (e.g., accumulator full).
Counterbalance Valve	Normally Closed (NC)	Inlet + Pilot	Prevents an overrunning load from pulling the actuator faster than intended. Controls lowering speed.

> **Key Difference:**
> • Relief Valve = NC, senses **inlet** pressure → protects system.
> • Reducing Valve = NO, senses **outlet** pressure → regulates branch pressure.

6.2 Flow Control Valves

Valve	Description	Application
Simple Restrictor (Needle Valve)	Adjustable orifice	Basic speed control (affected by load changes)
One-Way Flow Control	Restrictor + Check valve in parallel	Controls speed in one direction; free flow in reverse
Pressure-Compensated Flow Control	Restrictor + internal pressure compensator	Maintains constant flow regardless of load pressure variations

Temperature-Compensated Flow Control	Compensates for viscosity changes	Maintains flow accuracy across temperature range